

Generation Interconnection System Impact Study Report

For Queue Project Q18-1113

Madison 100 kV
January 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1113. The Interconnection Customer proposes a 141.6 MW natural gas combined cycle facility interconnecting at the Madison 100 kV substation in the Indiana area. The request was received on November 24, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 272 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,303,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Eldean – Monashee 100kV	E Methuen 500-230 kV Auto	181.8	136	5.3
Madison Substation – Pendleton 100kV line	Allegheny Energy Unit 3 4 & 5 345/1 kV XFMR	163.6	85	100.0
Carbondale Nw – Eldean 100kV L	East Methuen 500-230 kV Transformer Bank	161.6	165	5.3
Monashee – W Iverside 100kV line	E Methuen 500/230kV Transformer Bank	152.9	145	5.3
Carbondale Nw – Richard Lake 100kV L	E Methuen 500 kV/230 kV Transformer Bank	151.7	192	5.3
Savona – Richard Lake 100kV line	E Methuen 500 kV/230 kV TX	144.1	219	5.3
Savona – Butler Municipal Power Plant 100kV	E Methuen 500-230 kV Transformer	140.0	237	5.3
Butler Municipal Power Plant Substation – Hansel 100kV	E Methuen 500/230kV XFMR	137.0	253	5.3
Hansel Substation – Thousand Springs Substation 100kV	E Methuen 500 kV/230 kV Auto	135.1	263	5.3
Websterville Substation – Anchor Hocking 100kV Line	Anchor Hocking – Semass Tap 100kV	131.1	119	6.5
Cape N Substation – Anchor Hocking 100kV	East Methuen 500/230kV Transformer Bank	130.0	191	8.2
Websterville – Powerline Elkhorn 100kV Line	Murdock 100 kV/1 kV Transformer	129.6	134	5.2
Eaton – Cape North 100kV	E Methuen 500-230 kV Auto	128.5	200	8.2
Thousand Springs – Powerline Elkhorn 100kV	Mrdock 100/1 kV TX	122.4	127	5.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Semass Tap Substation – Snyder 100kV	Websterville – Powerline Elkhorn 100kV	114.6	162	5.3
Semas Tap Substation – Anchor Hocking 100kV	Powerline Elkhorn – Websterville 100kV	113.6	169	5.3
Pendleton – Anderson 100kV Line	Allegheny Energy Units 3 4 & 5 345/1kV XFMR	111.6	125	100.0
S Haven – Anderson Substation 100kV	Allegheny Energy Units 3 4 & 5 345/1kV Auto	105.1	300	100.0
Dutton Pump – Bourne 100kV	Hansel – Thousand Springs 100kV Line	100.8	115	5.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Eldean – Monashee 100kV	Rebuild line with higher-capacity conductor	\$3,490,000	10% (\$361,000)
Madison Substation – Pendleton 100kV line	Rebuild line with higher-capacity conductor	\$3,023,000	100% (\$3,023,000)
Carbondale Nw – Eldean 100kV L	Rebuild line with higher-capacity conductor	\$2,133,000	10% (\$221,000)
Monashee – W Iverside 100kV line	Rebuild line with higher-capacity conductor	\$2,587,000	10% (\$268,000)
Carbondale Nw – Richard Lake 100kV L	Rebuild line with higher-capacity conductor	\$3,409,000	10% (\$353,000)
Richard Lake – Savona 100kV	Rebuild line with higher-capacity conductor	\$10,098,000	10% (\$1,045,000)
Savona – Butler Municipal Power Plant Substation 100kV	Rebuild line with higher-capacity conductor	\$2,437,000	10% (\$252,000)
Butler Municipal Power Plant – Hansel 100kV line	Rebuild line with higher-capacity conductor	\$4,068,000	10% (\$421,000)
Hansel Substation – Thousand Springs Substation 100kV	Rebuild line with higher-capacity conductor	\$5,694,000	10% (\$590,000)
Websterville Substation – Anchor Hocking 100kV Line	Rebuild line with higher-capacity conductor	\$5,327,000	9% (\$467,000)
Cape N – Anchor Hocking 100kV line	Rebuild line with higher-capacity conductor	\$4,567,000	9% (\$412,000)
Websterville – Powerline Elkhorn 100kV Line	Rebuild line with higher-capacity conductor	\$16,441,000	15% (\$2,451,000)
Eaton – Cape N 100kV	Rebuild line with higher-capacity conductor	\$7,663,000	9% (\$691,000)
Thousand Springs – Powerline Elkhorn 100kV	Rebuild line with higher-capacity conductor	\$2,989,000	15% (\$446,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Snyder – Semass Tap 100kV L	Rebuild line with higher-capacity conductor	\$4,449,000	10% (\$434,000)
Semas Tap Substation – Anchor Hocking 100kV	Rebuild line with higher-capacity conductor	\$15,496,000	10% (\$1,513,000)
Pendleton – Anderson 100kV	Rebuild line with higher-capacity conductor	\$1,723,000	100% (\$1,723,000)
S Haven – Anderson Substation 100kV	Reconductor line	\$3,180,000	48% (\$1,534,000)
Dutton Pump – Bourne 100kV	Reconductor line	\$710,000	14% (\$98,000)
	Total allocated to project		\$16,303,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.